

PRATHAM SHAH

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EXPERIENCE

Cloud Security Reliability Engineer Intern / Searce Inc. Apr 2026 – Aug 2026

- Analyzed Google Cloud infrastructure utilization, billing data, and usage patterns to identify cost inefficiencies and recommend resource rightsizing opportunities.
- Developed and implemented cloud cost optimization strategies that helped a client achieve \$2,000+ in savings, balancing cost efficiency with application reliability and performance.

AI Full-Stack Engineering Intern (QA Focus) / NeuraMach AI Studios Jan 2026 – Apr 2026

- Designed and implemented a secure AWS video delivery solution using S3 and CloudFront, incorporating signed cookies, authentication-based access control, and protected content delivery.
- Resolved 80+ production defects and 20+ backend errors, improving application stability and reducing recurring issues in the existing codebase.
- Identified and remediated 6 security vulnerabilities, and refactored cluttered legacy code into structured, reusable React components to improve maintainability.

Frontend Developer / Abhinandan Mountreea (Freelance) Jul 2025 – Nov 2025

- Translated business and UX requirements into a high-performance real estate platform using React.js, Vite, and GSAP.
- Integrated Cloudinary CDN, WebP conversion, caching, and lazy loading, reducing initial page load time by 50% and increasing user engagement by 30%.

PROJECTS

InsightVault: LLM-Powered Semantic Search System (Python)

- Architected a semantic search and knowledge retrieval system powered by LLMs, with backend APIs optimized for caching and low latency.
- Integrated vector-based search to enable accurate, context-aware retrieval over unstructured content, with a scalable, modular backend architecture.

StreamVault: Cloud Video Streaming Platform (Python)

- Built a secure video streaming platform using AWS CloudFront, S3, and Cognito to deliver protected HLS video streams.
- Automated video transcoding using FFmpeg on EC2 Spot Instances, reducing video processing costs by over 90%.
- Developed RESTful APIs using FastAPI and DynamoDB to save and resume user playback progress.

Paradise Nursery: React E-Commerce App (TypeScript, React)

- Built a React 19 e-commerce application (React Router, Vite) for browsing and purchasing products across categorized listings.
- Implemented cart state management (add/remove items, quantity updates, real-time total cost calculation) using reusable components (ProductCard, CartItem, Header).

Live demo: paradise-nursery-inky.vercel.app

Source: github.com/pshah-lab/paradise-nursery

CERTIFICATIONS

- Google Cloud Associate Cloud Engineer (ACE)
- Claude Certified Architect - Foundations (CCA-F), Anthropic (In Progress)

EDUCATION

MIT Art, Design and Technology University Aug 2022 – Jun 2026

B.Tech in Computer Science

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript

Frontend: React.js, React Router, HTML5, CSS3, Responsive Web Design

Backend & Databases: Node.js, Express.js, FastAPI, REST APIs, MongoDB, PostgreSQL

Cloud & DevOps: AWS (EC2, S3, CloudFront), Google Cloud, Docker, Git, GitHub Actions

AI/ML: LLM Applications, RAG, Semantic Search, Vector Search, Prompt Engineering

ACHIEVEMENTS

- Second Runner-up at SHODH 1.0 Campus Problem Hackathon (2025) for an innovative full-stack solution.
- Published research paper on BCI-based robotic limb control in IJIRT.